UNIVERSITY OF SINDH

JAMSHORO SINDH, PAKISTAN

B.Ed. (04 Years)

Course No. GNAS– 301 **Credit Hours 3**

Title of Course ***ENVIRONMENTAL SCIENCE*** ***Semester*** ***1st***

Course Description

The course Environmental Science is designed to provide students with a comprehensive understanding of the interactions between humans and their environment, with a particular focus on environmental challenges, climate change, and their implications for human health and sustainability. It explores the structure and functioning of ecosystems, natural resource management, and major environmental issues at global, regional, and local levels, including those relevant to Pakistan.

The course emphasizes environmental pollution (air, water, and soil), waste management practices, occupational health and safety, and risk assessment techniques. It also introduces students to toxicology, environmental laws, international agreements, and sustainable development frameworks such as the Sustainable Development Goals (SDGs).

Through theoretical knowledge and practical insights, students will develop skills to analyze environmental problems, apply public health tools and methods, and propose effective solutions for environmental protection and health promotion. The course aims to foster environmental awareness, responsibility, and informed decision-making among future educators.

Course Oucome

Course Learning Outcomes (CLOs)

By the end of this course, students will be able to:

1. **Explain** fundamental concepts of environmental science, including ecosystems, natural resources, and human–environment interactions.
2. **Analyze** major environmental issues such as pollution, climate change, and resource depletion at local and global levels.
3. **Evaluate** environmental health risks and the effects of physical, chemical, and biological hazards on human health.
4. **Apply** environmental management tools and techniques for pollution control, waste management, and risk assessment.
5. **Develop** sustainable solutions and demonstrate responsible environmental behavior in personal and professional contexts.

Course Outline

Week 1

1.1 Introduction to Environmental Science, scope, importance

1.2 The overview of natural environment structure of planet earth

1.3 The nexus between environmental problems and anthropogenic activities
1.4 Global, regional and local environmental issues and their current status

Week 2

2.1 Introduction to ecosystems, types, importance, threats and conservation
2.2 Natural resources, types, services, issues and management
2.3 Individual responsivities to protect natural environment

Week 3

3.1 Global water distribution, water resources of Pakistan, water security and challenges
3.2 Water pollution, sources, types, impacts, and remedies
3.3 Industrial effluents, sources, impacts, treatment and recycling

Week 4

4.1 The principles of municipal solid waste management
4.2 Hazardous waste management: hospital and industrial waste

Week 5

5.1 Air pollution, causes, effects, and control strategies
5.2 Noise pollution, sources, human health impacts and control measures

Week 6

6.1 Health, safety and environment: Hazards
6.2 Physical, mechanical, biological, chemical and psychological hazards and stress
6.3 Health and environment, Environmental safety
6.4 Hazards identification and risk assessment and management process

Week 7

7.1 Mid Term Examination

Week 8

8.1 Workplace environment: Occupational safety and health issues in industries, construction, agriculture and service sectors
8.2 Accidents, injuries and workplace fatalities
8.3 Occupational safety and health management systems (international and national)

Week 9

9.1 Safety Management: Regulations of health, safety and environment
9.2 Internal control and management philosophy
9.3 Duties and rights, strategies and goals
9.4 Roles and responsibilities of occupational health and safety professionals

Week 10

10.1 Key principles of management and HSE
10.2 Measures and models for HSE
10.3 Organizational environment
10.4 HSE statutes and regulations
10.5 Establishing HSE plans
10.6 Challenges of health within working environment
10.7 External environment and safety
10.8 Different tools and instruments

Week 11

11.1 Toxic impacts of atmospheric agents
11.2 Fate of absorbed toxins and xenobiotics, including detoxification and bioactivation
11.3 Natural detoxification processes
11.4 Risk management

Week 12

12.1 Multilateral environmental agreements
12.2 Sustainable development goals (SDGs)
12.3 Environmental laws, policies, guidelines, strategies related to public health

Week 13

13.1 Introduction to Toxicology
13.2 Classification and properties of toxic substances
13.3 Anthropogenic and natural poisons
13.4 Acute and chronic effects
13.5 Genotoxic, mutagens, teratogens, carcinogens and sensitizers

Week 14

14.1 Biological properties of organic and inorganic pollutants
14.2 Essentiality and toxicity
14.3 Routes of absorption
14.4 Bioaccumulation and bio-magnificatio

Week 15

15.1 Greenhouse gas effect, global warming, causes, effects
15.2 Smog, types, causes, human health effects, control measures

Week 16

16.1 Climate Change, causes, effects, mitigation strategies and adaptation
16.2 Final Term Examination

Teaching Strategy

Interactive lectures, seminars, group work, case-studies, presentations, guided reading, home study and individual tutorials will be used to develop students' knowledge and understanding of Environmental Science. Theories and concepts necessary for the course will be introduced.
Students will be encouraged to read widely and to keep abreast of current trends and issues.
Discussions and presentations will be used to develop course themes and to facilitate the sharing of ideas and knowledge.

ASSESSMENT CRITERIA

Mid-Term Marks	Final Term Marks	Assignment/ Presentation Marks	Attendance Marks	Total
30	50	10	10	100

Recommended Books

1. Oxford Textbook of Environmental Science, Vol. I & II by Walter W. Holland
2. Planetary Health – Andy Haines and Howard Frumkin
3. Environmental Health Practice – RSF Shilling
4. Occupational Health – J.M. Harrington / F.S. Gill
5. Environmental Studies – Clark and Henderson
6. Environmental Science in South East Asia – WO Phoon, PCY Chen
7. Textbook of Preventive of Pollution – J.E. Park, K. Park